Flutter on OpenHarmony: A New Embedder Direction

Linaro Solutions & Services Group
VP & GM, Linaro Solutions & Services
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The Challenge

Architecture Mismatch

Flutter's architecture mismatch with OpenHarmony

Key incompatibilities:

✓ Impeller/Vulkan graphics pipeline ≠ Cangjie rendering

✓ Android embedder ≠ ArkUI framework

✓ APK packaging ≠ HAP packaging

Maintenance Burden

Current implementation requires adapting all patches to Flutter:

→ Every new Flutter engine release requires rework

→ Slows adoption of new Flutter features dramatically

→ Increases maintenance burden significantly

Three Approaches Evaluated

Approach	Description	Effort	Risk
A	Full native embedder rewrite	Very High	High
B	Fork Impeller for OHOS	High	Medium
C	Flutter Web engine + OpenHarmony Embedder	Medium	Low

Recommended Solution (Direction C)

Approach Overview

Flutter Web Engine → compiled to Cangjie

OpenHarmony Flutter Embedder

Native API Bridge → OHOS Capabilities

Industry Trend — Everyone Is Building Embedders

Google: Android team actively migrating to Flutter's embedder API

→ GitHub issue #176649: "Migrate Android to embedder API"

→ Reference: https://github.com/flutter/flutter/issues/176649

Sony: flutter-embedded-linux (1,300+ stars)

→ Embedded Linux devices, smart displays, automotive

Samsung: Tizen-based Flutter initiatives

→ IoT and wearable devices

Why Direction C Wins

- ✓ Maximum code reuse (Flutter web engine, widgets, tooling)
- ✓ Respects OpenHarmony's native architecture
- ✓ Production-ready path with manageable scope
- ✓ Leverages existing Flutter ecosystem
- ✓ Google actively developing web engine — long-term viability

Phase 1 Results (Completed April 2025)

Feasibility validation: ✓ Confirmed

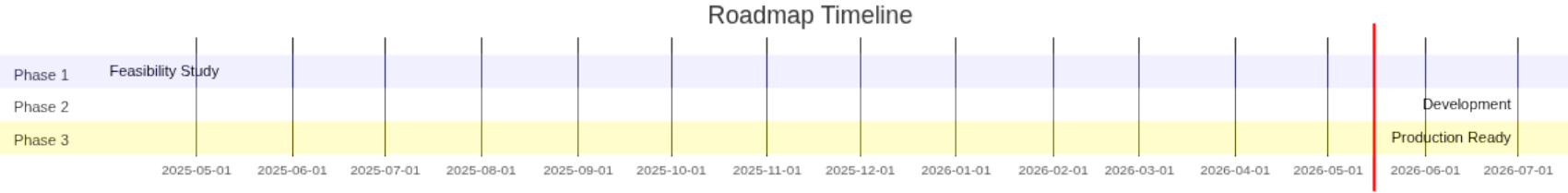
Cangjie compilation target: ✓ Working

Web engine performance: ✓ Acceptable for UX

Key findings: Technical approach is sound

Roadmap

Phase	Timeline	Focus
Phase 1	Apr 2025	Feasibility study (COMPLETE)
Phase 2	Q3 2026	Development with Flutter community
Phase 3	TBD	Production-ready features



Applications Migration Considerations

Concurrent Activities During Phase 2:

Performance Evaluation: New architecture vs current implementation

Migration Cost Analysis: Identify and minimize costs for existing apps

Tooling Development: Build migration tool and strategy

New App Development: Develop new apps on new architecture

Migration Benchmarks: Use old apps as benchmarks

Migration Strategy:

→ New apps → Direct to new architecture (avoid future migration)

→ Existing apps → Migrate progressively as benchmarks

→ Risk mitigation through early challenge identification

Migration Flow:

Linaro's Role — Community Bridge

Connecting Communities

✓ Open-source communities to device makers
✓ Technical expertise in both Flutter and OHOS ecosystems
✓ Bringing Flutter community knowledge to OHOS adaptation

Influence Direction

→ Influencing direction for production readiness

Linaro's Role — Industry Mediator

Supporting Huawei Divisions

- ✓ Consumer, Compute, Cloud, Networking
- ✓ Other OpenHarmony-adopting OEMs

Services Offered

- → Adaptation and development
- → Testing and compliance (CRA, ISO 27001)
- → Deployment and long-term support

Thank You + Q&A

Contact information Linaro website Link to presentation materials Questions welcome